Cross-Asset Volatility & Shock Regime Framework

WTI Crude Oil | Natural Gas | S&P 500

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KEY MESSAGE

Markets are currently best characterized by a regime-switching volatility structure where:

- *Energy markets lead systemic volatility expansion*
- *Correlations compress during stress periods*
- *Shock events cluster rather than occur independently*
- *Diversification benefits are state-dependent, not stable*

Implication:

Risk must be managed dynamically across regimes — static correlation and volatility assumptions materially understate tail exposure.

1. MARKET STRUCTURE OVERVIEW

This framework evaluates volatility regimes and shock behavior across:

- *WTI Crude Oil*
- *Natural Gas*
- *S&P 500*

The objective is to identify how risk transmission evolves across assets under changing volatility regimes.

Markets are segmented into:

- *Calm*
- *Normal*
- *Stress*

with additional identification of shock events (2× rolling volatility threshold).

2. CORE MARKET FINDINGS

- *Energy assets exhibit persistent volatility clustering*
- *Natural gas shows highest shock frequency across all assets*

- *Stress regimes drive cross-asset correlation convergence*
- *Equity markets act as a secondary transmission channel during risk-off periods*

EXHIBIT 1 — CROSS-ASSET CORRELATION STRUCTURE

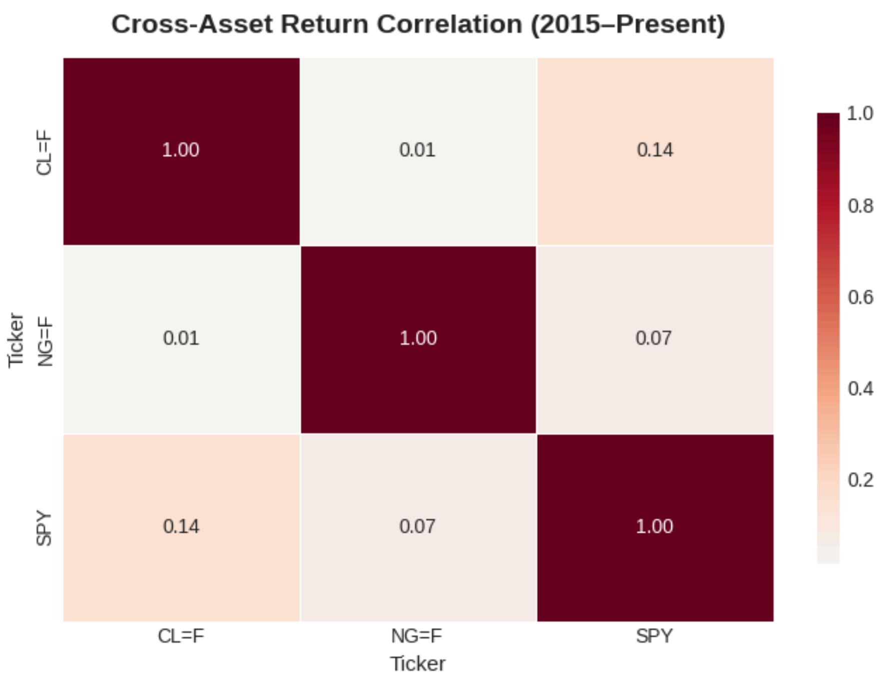

Figure 1: Correlation Matrix (WTI | NG | SPY)

Key Observation:

- *Correlation increases significantly during stress regimes*
- *Diversification breakdown is most pronounced in energy-equity linkages*

Market Implication:

> *Portfolio hedges weaken precisely when they are most needed.*

EXHIBIT 2 — WTI VOLATILITY REGIME STRUCTURE

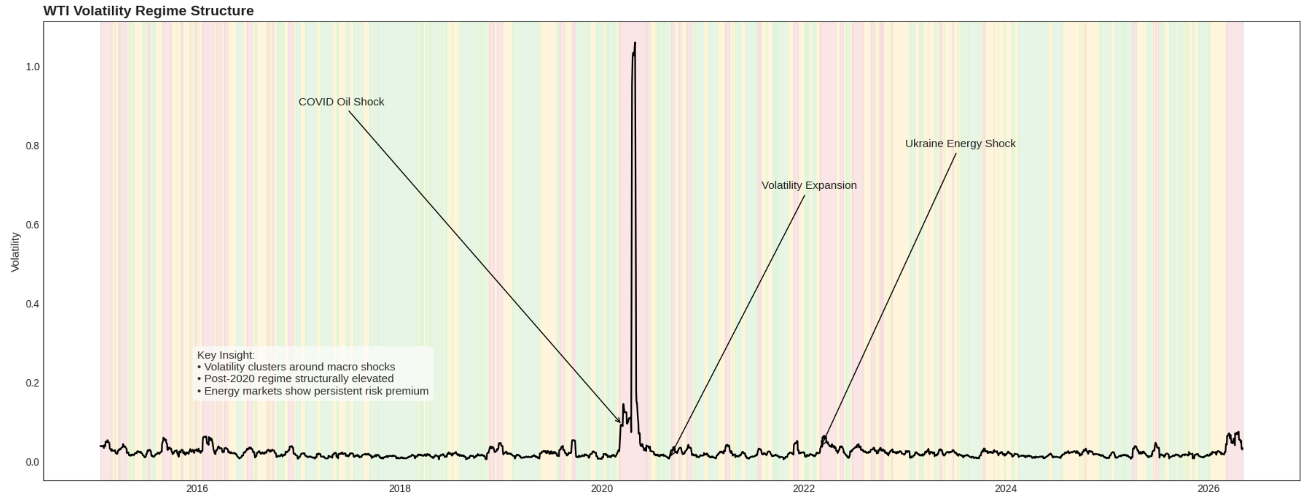

Figure 2: WTI 10-Day Rolling Volatility (Regime Shaded)

Green = Calm | Yellow = Normal | Red = Stress

Key Observation:

- *Volatility clustering is persistent across multi-year horizon*
- *Post-2020 baseline volatility remains structurally elevated*

Market Implication: *Historical volatility underestimates forward energy risk.*

EXHIBIT 3 — SHOCK EVENT DISTRIBUTION

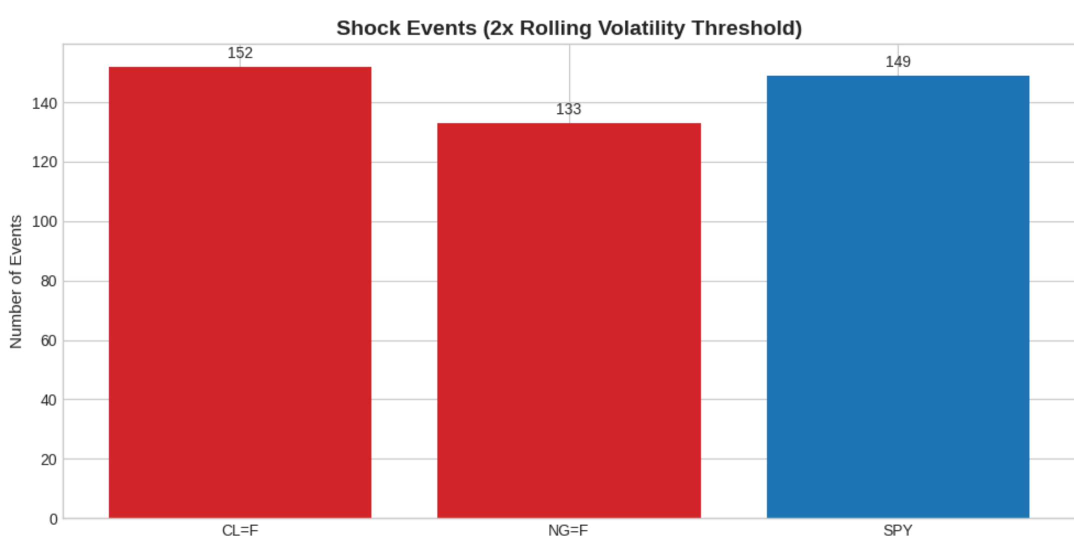

Figure 3: Shock Frequency (2× Volatility Threshold)

Key Observation:

- *Natural gas dominates shock frequency distribution*
- *Shocks are clustered, not isolated*

Market Implication:

Gas behaves as a dislocation-driven market rather than a smooth financial time series.

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3. REGIME INTERPRETATION

Market behavior is best understood as a state-dependent system:

Calm Regime

- *Low volatility*
- *Stable correlations*
- *Low systemic risk transmission*

Stress Regime

- *Correlation convergence*
- *Volatility expansion*
- *Cross-asset risk transmission*

Structural Insight:

Risk is transmitted through regime transitions, not linear price evolution.

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4. TRADING & RISK IMPLICATIONS

1. Dynamic Correlation Risk

- *Correlation is regime-dependent and rises materially during stress periods.*

2. Energy-Led Volatility Transmission

- *WTI and natural gas act as leading indicators of systemic risk expansion.*

3. Shock Clustering Risk

- *Risk events are clustered → tail risk is path-dependent, not isolated.*

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5. CONCLUSION

This framework demonstrates that:

- *Volatility is regime-based, not constant*
- *Correlations are conditional on stress states*
- *Energy markets lead systemic risk transmission*
- *Shock behavior is clustered and non-random*

FINAL TAKEAWAY

Markets operate as a regime-switching risk system where stress conditions dominate correlation structure and materially reduce diversification effectiveness.